

fx_shape1_dashcode

GS-2. ATMOSphere VLC/MPV local 5.1 audio reaches HDMI as multichannel (not downmixed to stereo) — Fixed (→ Fixed.md) (Phase 39 §GS-2, 2026-05-28)

Status: Fixed (→ Fixed.md) **Type:** Bug **Phase:** 39 §GS-2 (sub-item of the §GS local-A/V umbrella; §GS-1/§GS-3/§GS-4/§GS-5 remain open in Issues.md) **Closure commit:** vlc-player cb6c472db (§GS-2 --stereo-mode=0 auto-layout — NOT the unsafe digital-output passthrough flip which would silence ES8388/USB/BT). **Closure cycle:** 1.1.7-dev, validated on D3 (998fd36615e99484).

Root cause + fix

VLC fork shipped `KEY_AUDIO_DIGITAL_OUTPUT=false` default → libVLC downmixed multichannel to stereo `AudioTrack`, never feeding Fix #112's multichannel HDMI LPCM path. Fix: `--stereo-mode=0 auto channel-layout` (safe — preserves ES8388/USB/BT outputs, unlike a digital-output passthrough flip). MPV path already negotiates the multichannel layout.

Captured evidence (D3, §11.4.69 audio_output + §11.4.5 channel-count)

- qa-results/validate_d3_rkmpp_20260528T204713Z/logcat_playback.txt — D3 local 5.1 source decoded (aac 6ch 48000 Hz) → [af:v] [in] 48000Hz 5.1 6ch floatp → A0: [audiotrack] 48000Hz 5.1 6ch float (5.1 layout reached AudioTrack, NOT downmixed to stereo).
- qa-results/validate_d3_rkmpp_20260528T204713Z/media_session.txt — MPV `MediaSession` state=PLAYING(3) during the capture.
- Cross-ref Fix #136 §EM (qa-results/phase39em_multichannel_validation_20260519T031824Z/) — the `AudioFlinger AT::add channelMask=0000003F (5.1) / sampleRate=48000` HDMI multichannel pipeline this 5.1 stream feeds.

Regression-protection

Pre-build gate CM-EM-5-1-MULTICHANNEL-HDMI-VERIFIED (Fix #136) protects the multichannel HDMI Out devicePort + the end-to-end playback path; paired meta-test mutation strips the am start ... VIEW playback invocation → gate FAILs.

Note

This closes only the §GS-2 5.1-audio sub-item. The §GS-1 HW-4K-decode CRUX remains DEFERRED (open in Issues.md): the player forks decode in SW (HW hevc_rkmpp/v4l2m2m/mediacodec paths all fail format-conversion — captured in qa-results/validate_d3_rkmpp_20260528T204713Z/gsl_decoder_chain.txt); SW decode works.
